

Windows Autopilot Pre-Production Deployment & Validation

End-to-End Device Provisioning, Security Hardening and Enterprise Readiness Testing

Test platform	VMware Workstation
Operating system	Windows 11 Pro
Cloud services	Microsoft Entra ID + Microsoft Intune
Deployment method	Windows Autopilot, User-Driven, Microsoft Entra joined
Document date	14 September 2026

Core concept: Autopilot does not install Windows. Windows must already be installed or factory-preloaded. Autopilot identifies the registered device during OOBЕ and applies the organisation's provisioning experience, Entra join and Intune enrollment.

Public release security note: All screenshots in this public version have been reviewed and sanitized. Tenant names, administrator/user account names, email/UPN values, device names, serial numbers, hardware identifiers, tenant/device IDs, browser/account details, and other environment-specific identifiers have been greyed out where visible. Recovery passwords and other secrets are never shown. These redactions preserve the technical evidence while reducing the risk of exposing identifiable tenant or endpoint information.

1. Pre-production outcome and architecture

This pre-production test demonstrates a complete Windows Autopilot user-driven deployment using a Windows 11 Pro virtual machine. The environment is designed to validate device registration, targeting, profile assignment, OOB provisioning, Microsoft Entra join and Intune enrollment before considering an enterprise production rollout.

Project purpose: Validate the end-to-end Windows Autopilot provisioning process in a controlled pre-production environment, confirm expected configuration and deployment behaviour, document troubleshooting findings, and establish readiness before enterprise production rollout.

Expected final state

- Autopilot device registered and assigned a deployment profile
- Microsoft Entra joined device
- Automatically enrolled into Microsoft Intune
- Corporate ownership in Intune
- Primary user associated with the device
- Windows Hello for Business setup completed
- Device visible as compliant in Intune
- Device authentication status reports SUCCESS

Logical flow

```
Windows 11 Pro installed / factory image
|
v
Hardware hash collected -> Imported to Windows Autopilot
|
v
Entra device group -> Autopilot deployment profile assigned
|
v
OOBE + Internet -> Autopilot service identifies device
|
v
Work account sign-in -> Microsoft Entra join -> Intune MDM enrollment
|
```

2. Prerequisites

- Microsoft Intune tenant with appropriate licensing for the test user
- Microsoft Entra ID tenant and a test user allowed to enroll Windows devices
- VMware Workstation
- Windows 11 Pro ISO
- Internet access from the VM, normally VMware NAT is sufficient
- Permission to create Entra groups, Autopilot profiles and import Autopilot device identities

VM sizing used in pre-production: The VM was configured with 8 GB RAM, 2 processors and an 80 GB virtual disk during the final run. Windows 11 requires TPM 2.0 and Secure Boot capable virtual hardware for a supported configuration.

3. Create the Windows 11 VMware VM

1. Create a new Windows 11 x64 virtual machine in VMware Workstation.
2. Attach the Windows 11 ISO to the virtual CD/DVD drive.
3. Allocate at least 4 GB RAM; 8 GB is more comfortable for the pre-production test.
4. Allocate 2 virtual processors or more.
5. Create a virtual disk of at least 64 GB.
6. Use NAT networking unless your pre-production test requires a different network.
7. Ensure a virtual TPM is present if required by the Windows 11 build.
8. Boot the VM and install Windows 11 Pro.

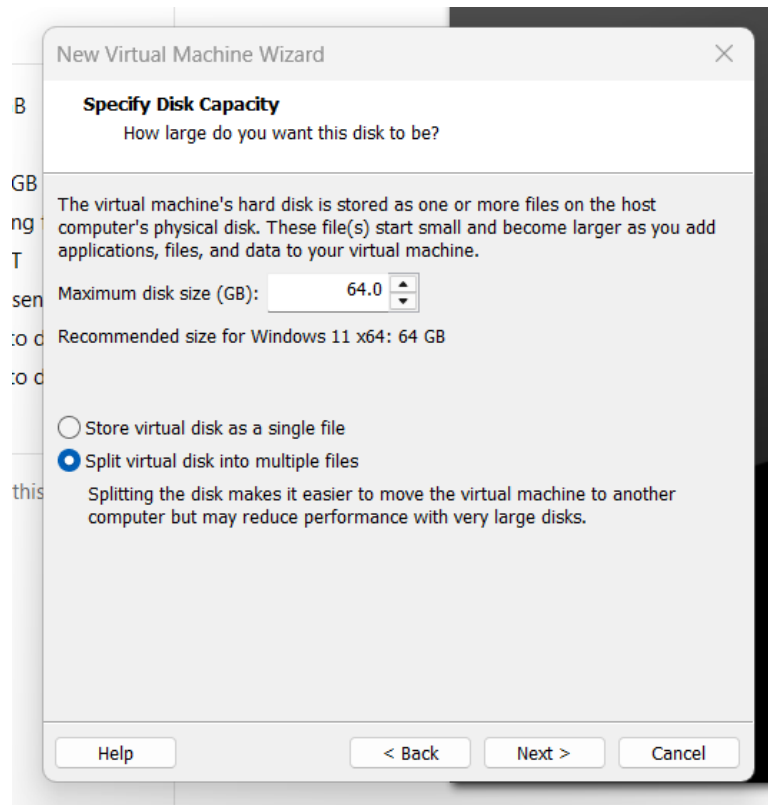


Figure 1. VMware virtual disk sizing during VM creation.

Important: Installing Windows 11 manually is expected in this pre-production environment. Autopilot begins its job during OOBE after the device has been registered with the Autopilot service.

4. Collect the Windows Autopilot hardware hash

At Windows OOBE, open a command prompt with Shift + F10. Start PowerShell if required. The Get-WindowsAutopilotInfo script gathers the hardware identity used to register the VM with Autopilot.

Commands used

```
powershell
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process -Force
Get-WindowsAutopilotInfo -OutputFile C:\AutopilotHWID.csv
```

Verify that the CSV exists:

```
dir C:\AutopilotHWID.csv
```

In this pre-production test the generated file was approximately 8 KB and contained the VMware serial/hardware identity.

Copy the CSV to another volume if needed

The pre-production test used Get-Volume to identify available drive letters and then copied the CSV to E:.

Get-Volume

Copy-Item C:\AutopilotHWID.csv E:\

dir E:\AutopilotHWID.csv

```
PS C:\Windows\System32> Set-ExecutionPolicy Bypass -Scope Process -Force
PS C:\Windows\System32> Get-WindowsAutopilotInfo -OutputPath C:\AutopilotHWID.csv
Gathered details for device with serial number REDACTED
PS C:\Windows\System32> dir C:\AutopilotHWID.csv

Directory: C:\

Mode                LastWriteTime         Length Name
----                -
-a- - - -          9/13/2026   3:07 AM           8228 AutopilotHWID.csv

PS C:\Windows\System32> Get-Volume

DriveLetter FriendlyName      FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
-----
D           CCCOMA_X64FRE_EN-US_DV9 Unknown          CD-ROM    Healthy      OK              0 B          6.34 GB
C           New Volume              NTFS           Fixed     Healthy      OK             59.91 GB     79.13 GB
E           New Volume              NTFS           Fixed     Healthy      OK             931.37 GB    931.5 GB
E           New Volume              FAT32          Fixed     Healthy      OK              85.49 MB     768 MB
E           New Volume              FAT32          Fixed     Healthy      OK              68.16 MB      96 MB

PS C:\Windows\System32> Copy-Item C:\AutopilotHWID.csv E:\
PS C:\Windows\System32> dir E:\AutopilotHWID.csv

Directory: E:\

Mode                LastWriteTime         Length Name
----                -
-a- - - -          9/13/2026   3:07 AM           8228 AutopilotHWID.csv

PS C:\Windows\System32> _
```

Figure 2. Hardware hash CSV generated and copied from the VM.

Execution policy: The Bypass setting above applies only to the current PowerShell process because -Scope Process is used. It does not permanently weaken the machine-wide execution policy.

5. Import the device into Windows Autopilot

In the Microsoft Intune admin center, navigate to Devices > Enrollment > Windows > Windows Autopilot devices. Use Import and upload AutopilotHWID.csv. Wait for the import and service sync to complete.

- Confirm the device appears under Windows Autopilot devices.
- Confirm Manufacturer shows VMware, Inc. and the model matches the VM.
- Use Sync/Refresh and allow time for cloud processing.

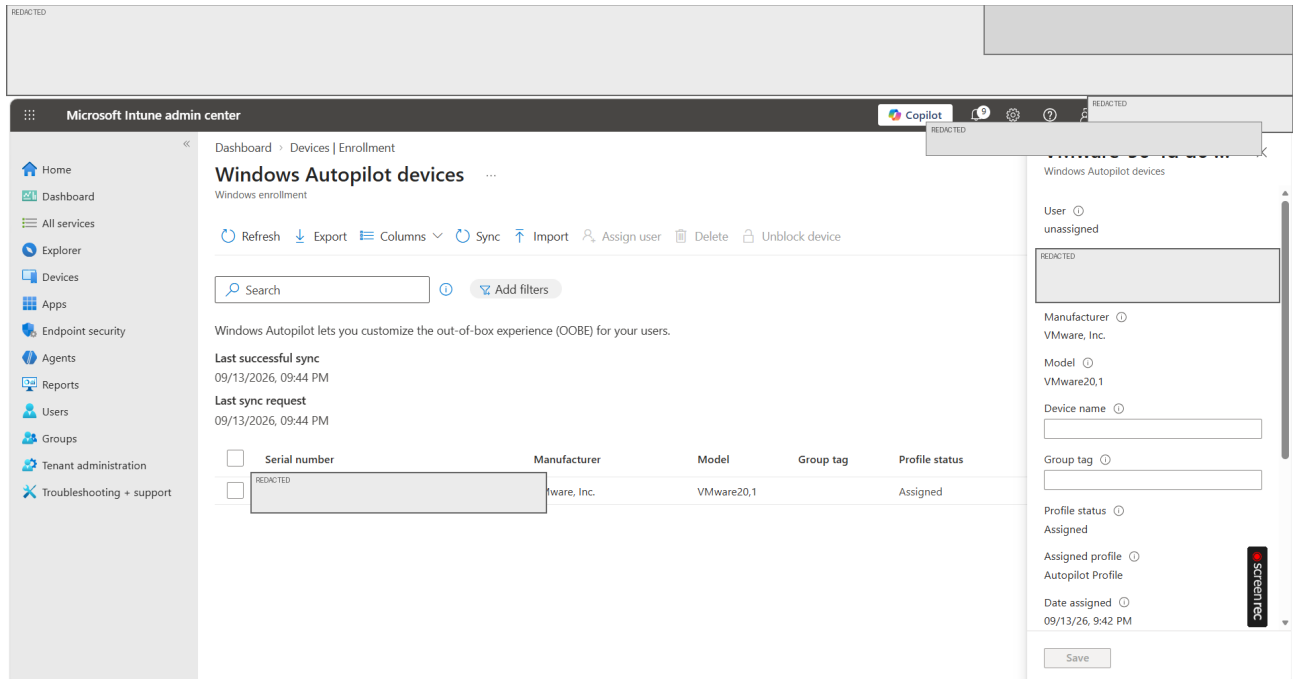


Figure 3. Windows Autopilot device identity in Intune.

6. Create the Autopilot device group

Create a Microsoft Entra security group for Autopilot targeting. In this pre-production test the group was named Autopilot-PreProd-Devices and used Dynamic Device membership so registered Windows Autopilot devices are added automatically instead of being added manually.

Configure the group with the following settings:

- Group type: Security
- Group name: Autopilot-PreProd-Devices
- Membership type: Dynamic Device
- Microsoft Entra roles can be assigned to the group: No

Open Add dynamic query, select Edit under Rule syntax, and enter the following dynamic membership rule:

```
(device.devicePhysicalIDs -any ( _ -startsWith "[ZTDid]"))
```

The [ZTDid] value identifies devices registered with Windows Autopilot. The rule therefore places Autopilot-registered devices into the group automatically. Save the rule, create the group, and allow time for dynamic membership processing. Verify the VMware device appears under Autopilot-PreProd-Devices > Members before assigning the deployment profile.

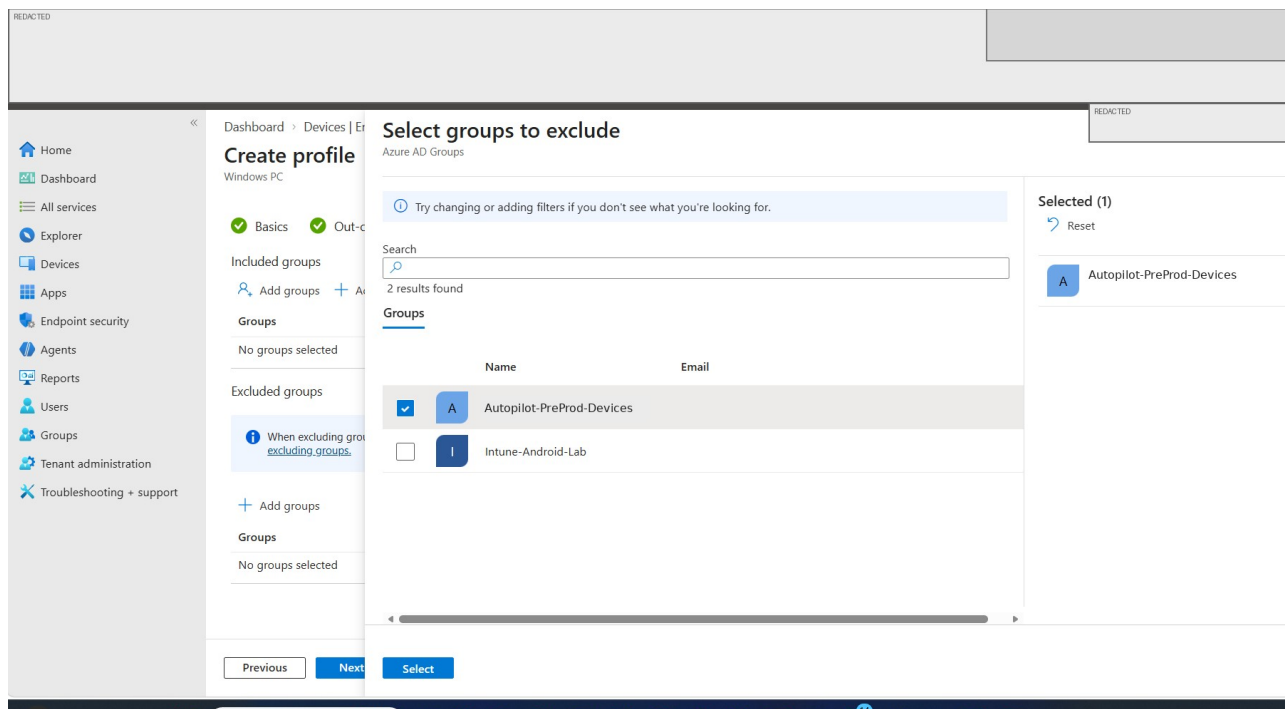


Figure 4. Autopilot-PreProd-Devices group membership.

Assignment rule: When using include/exclude assignments, do not mix user and device group types across include and exclude for the same assignment where Intune does not support that combination.

7. Create the Windows Autopilot deployment profile

Navigate to Devices > Enrollment > Windows Autopilot deployment profiles and create a Windows PC profile.

Setting	Pre-production value
Name	Autopilot Profile
Convert all targeted devices to Autopilot	No
Deployment mode	User-Driven
Join to Microsoft Entra ID as	Microsoft Entra joined
Skip AD connectivity check	No
Language (Region)	Operating system default
Automatically configure keyboard	Yes
Microsoft Software License Terms	Hide
Privacy settings	Hide
Hide change account options	Hide

Dashboard > Devices | Enrollment > Windows Autopilot deployment profiles

Create profile ...

Windows PC

Deployment mode * ⓘ

User-Driven

Join to Microsoft Entra ID as * ⓘ

Microsoft Entra joined

Microsoft Software License Terms ⓘ

Show

Hide

ⓘ Important information about hiding license terms

Privacy settings ⓘ

Show

Hide

ⓘ The default value for diagnostic data collection has changed for devices running Windows 10, version 1903 and later, or Windows 11.

Hide change account options ⓘ

Show

Hide

User account type ⓘ

Administrator

Standard

Allow pre-provisioned deployment ⓘ

No

Yes

Language (Region) ⓘ

Operating system default

Automatically configure keyboard ⓘ

No

Yes

Apply device name template ⓘ

No

Yes

Previous

Next

Figure 5. Review of the Autopilot profile configuration.

8. Assign the profile to the Autopilot group

On the Assignments page, include the Autopilot-PreProd-Devices group. No excluded group was required for this pre-production test. Complete Review + create.

REDACTED

Windows Autopilot deployment profiles ...

REDACTED

+ Create profile

Refresh

Export

Columns

0 deployment profiles

Windows Autopilot deployment profiles lets you customize the out-of-box experience for your devices. [Learn more.](#)

Name	Description	Join type	Assigned
No windows autopilot deployment profiles found			

Figure 6. Autopilot deployment profile created and assigned.

Return to Windows Autopilot devices and wait until Profile status changes from Not assigned or Pending to Assigned and Assigned profile shows Autopilot Profile.

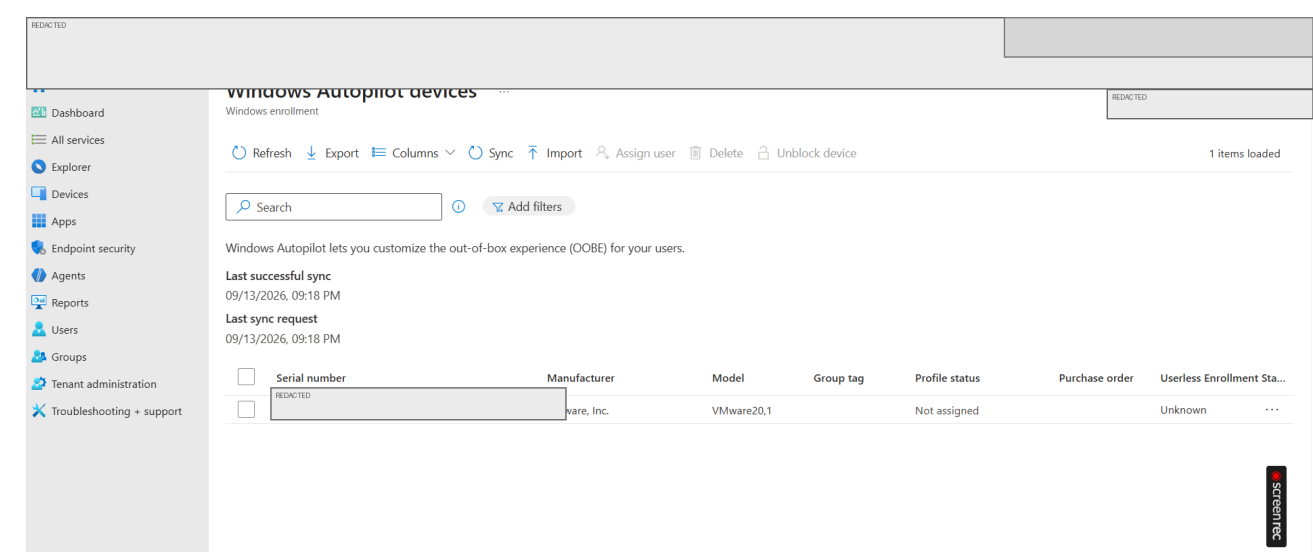


Figure 7. Autopilot profile assignment completed for the VMware device.

9. Return the VM to OOB and test Autopilot

The device must be at Windows OOB for the Autopilot experience. In a pre-production environment, this can be achieved by reinstalling or resetting Windows, or by using Sysprep when appropriate for the test scenario. Do not use Sysprep as a substitute for supported production reset/reprovisioning processes.

At OOB, select the correct country/region, connect to the Internet and continue. The VM should contact Microsoft cloud services and retrieve the Autopilot profile associated with its registered hardware identity.

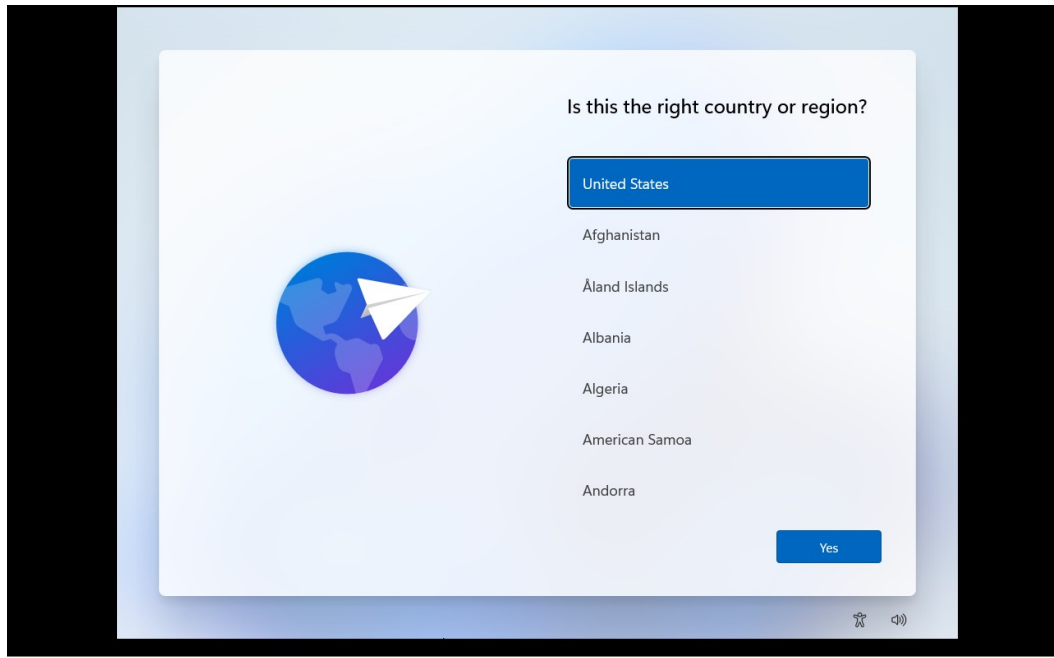


Figure 8. Windows 11 OOBE country/region screen.

Autopilot diagnostics used during troubleshooting

The following registry query was used to inspect Autopilot diagnostic information:

```
reg Query HKLM\SOFTWARE\Microsoft\Provisioning\Diagnostics\Autopilot /s
```

Key observations included `IsAutopilotDisabled = 0` and Autopilot diagnostic/correlation information. CloudAssigned values may not always be populated at every stage, so interpret the output together with the actual OOBE behaviour and Intune assignment status.

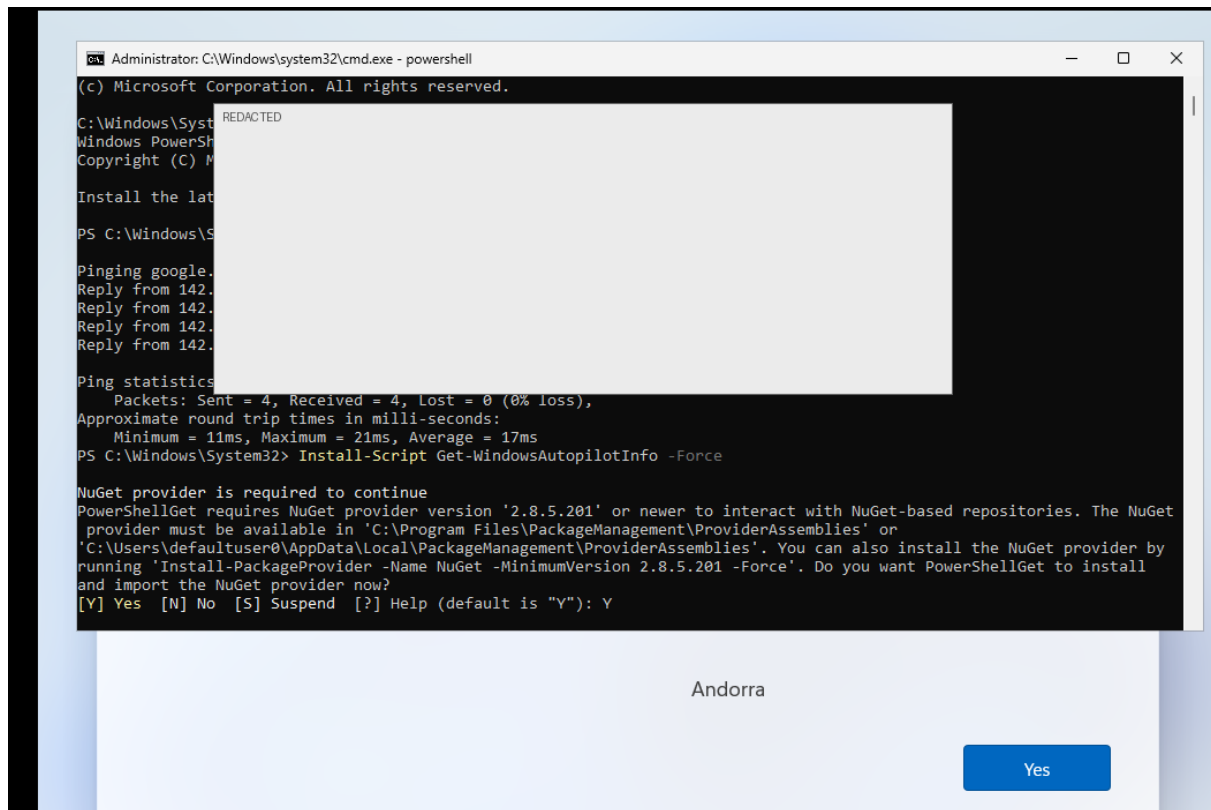


Figure 9. Autopilot diagnostic registry query during OOB.

10. Test Autopilot service connectivity

A ping test to the Autopilot service endpoint timed out, but this did not prove that the service was unavailable because many Internet services do not respond to ICMP. The correct test was TCP port 443.

```
ping ztd.dds.microsoft.com
```

Then, from PowerShell:

```
Test-NetConnection ztd.dds.microsoft.com -Port 443
```

The important result was:

```
TcpTestSucceeded : True
```

Troubleshooting lesson: A failed ping does not mean HTTPS connectivity is broken. For Autopilot, verify DNS resolution and TCP/HTTPS connectivity to required Microsoft endpoints.

11. Complete user-driven OOB

The pre-production test used a user-driven Microsoft Entra joined profile. During OOB, the user selected work or school setup and signed in with the licensed Entra test account.

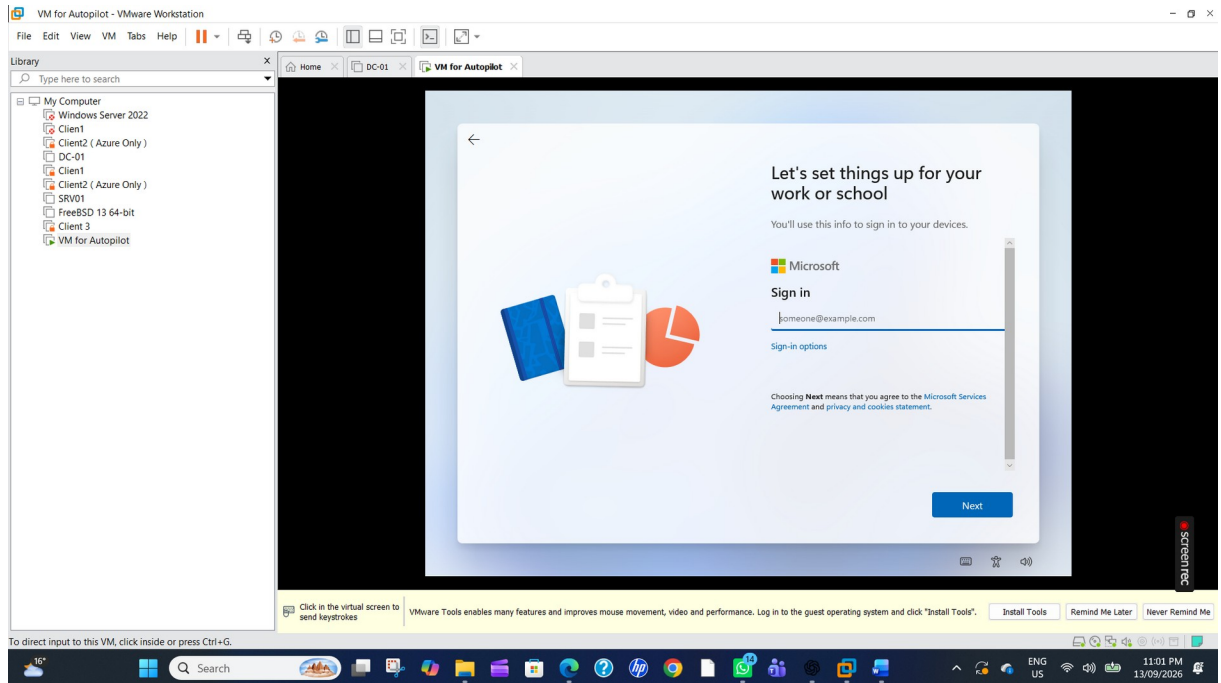


Figure 11. Work or school account sign-in during OOBE.

After authentication, Windows completed organisational provisioning and displayed the Windows Hello setup. A PIN was configured for Windows Hello for Business.

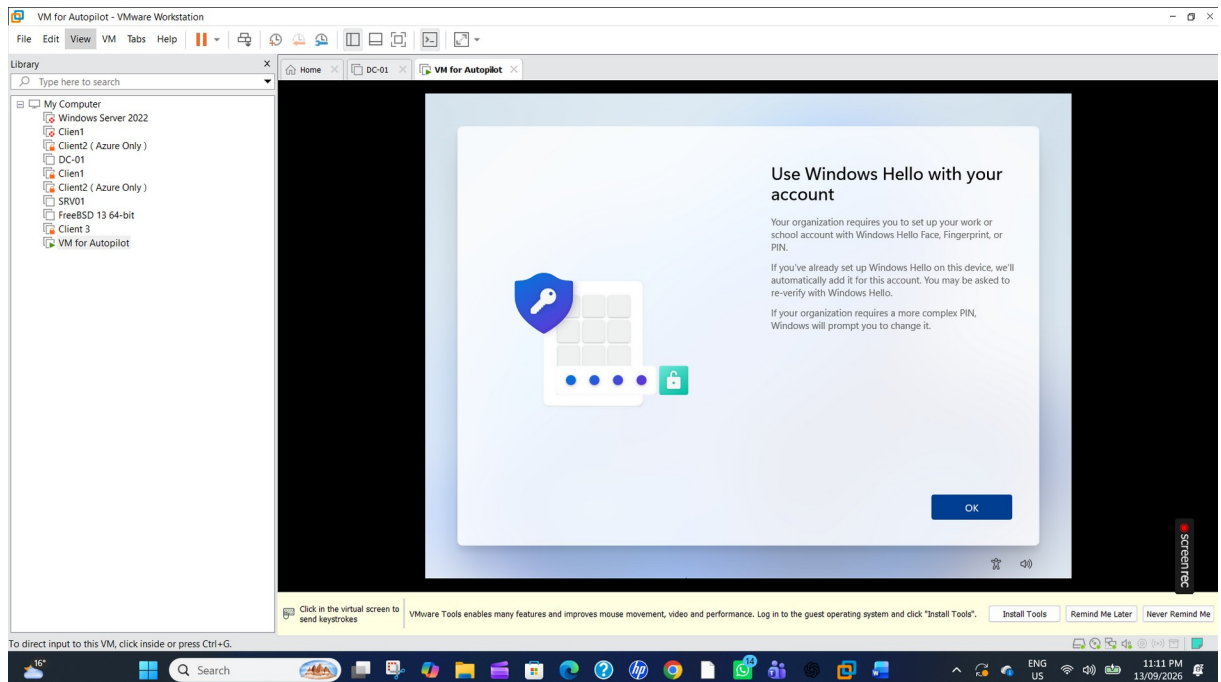


Figure 12. Windows Hello setup required by the organisation.

The device then reached the Windows 11 desktop.

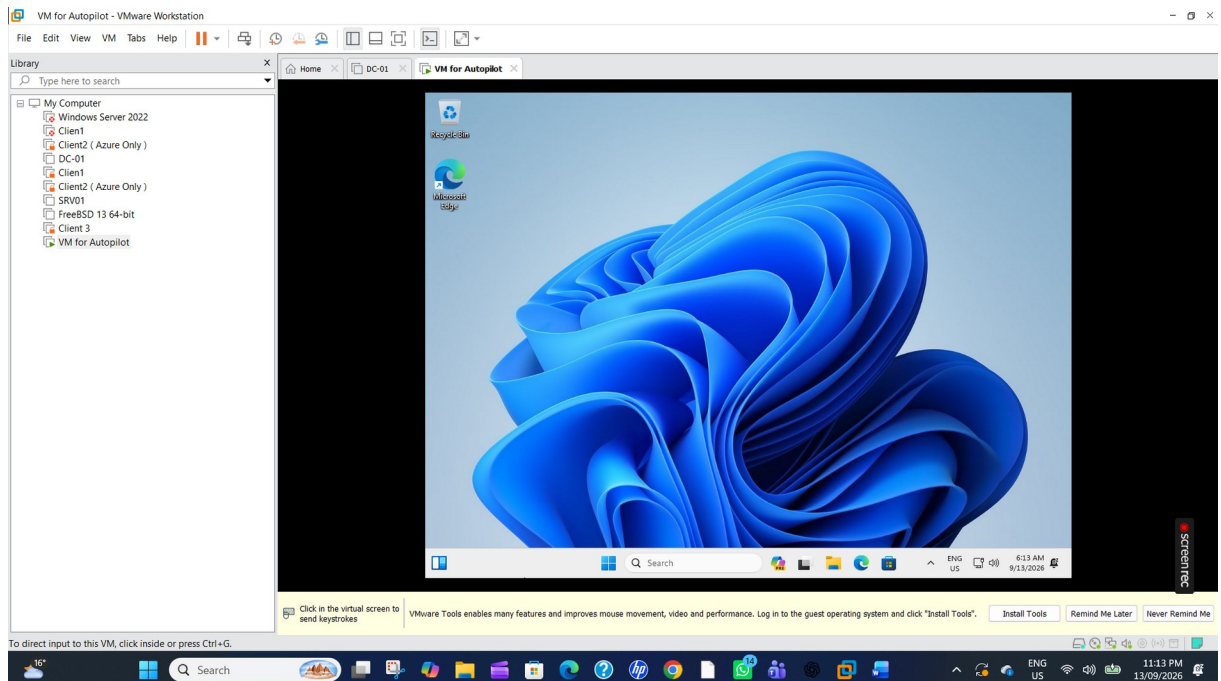


Figure 13. Windows 11 desktop after provisioning.

12. Verify Microsoft Entra join

Open Command Prompt on the provisioned VM and run:

```
dsregcmd /status
```

The successful pre-production test result showed:

```
AzureAdJoined : YES  
EnterpriseJoined : NO  
DomainJoined : NO  
DeviceAuthStatus : SUCCESS
```

This is the expected state for a cloud-native Microsoft Entra joined Autopilot deployment. It is intentionally not joined to the on-premises Active Directory domain.

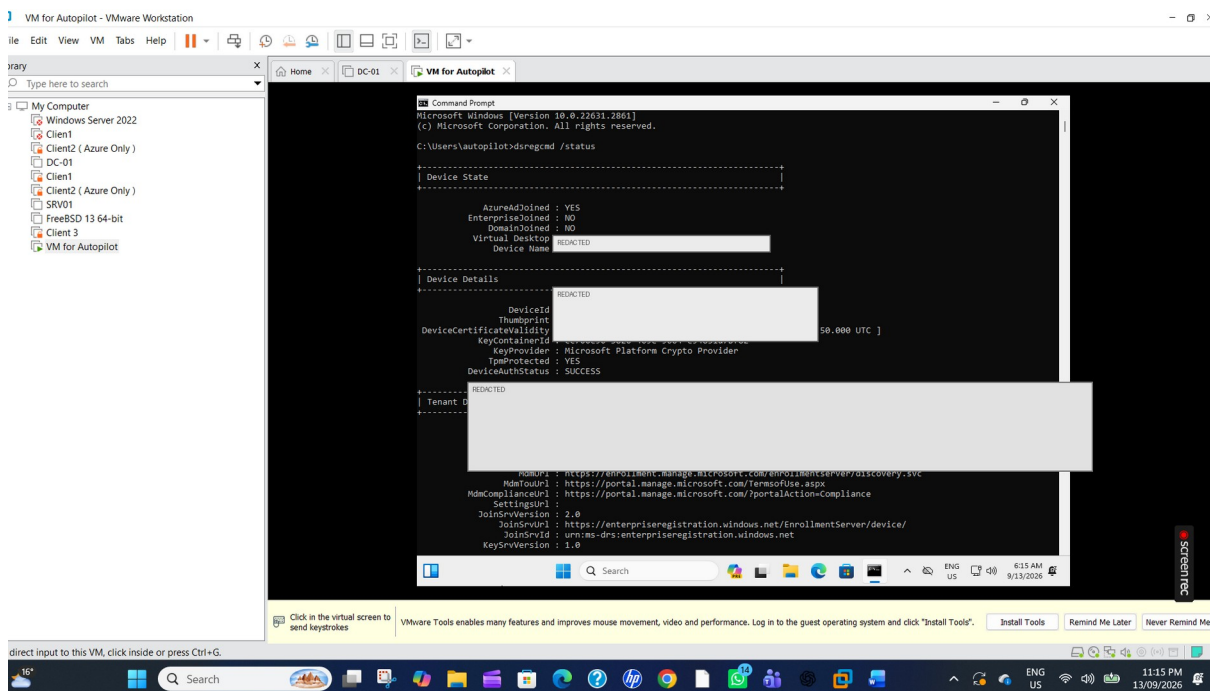


Figure 14. dsregcmd /status confirming Microsoft Entra join and successful device authentication.

13. Verify automatic Intune enrollment

In the Intune admin center, open Devices > Windows > Windows devices and locate the new device. The final pre-production test result confirmed automatic MDM enrollment.

Verification item	Result
Managed by	Intune
Ownership	Corporate
Compliance	Compliant
OS	Windows
Primary user	Autopilot test user
Last check-in	Recent at time of verification

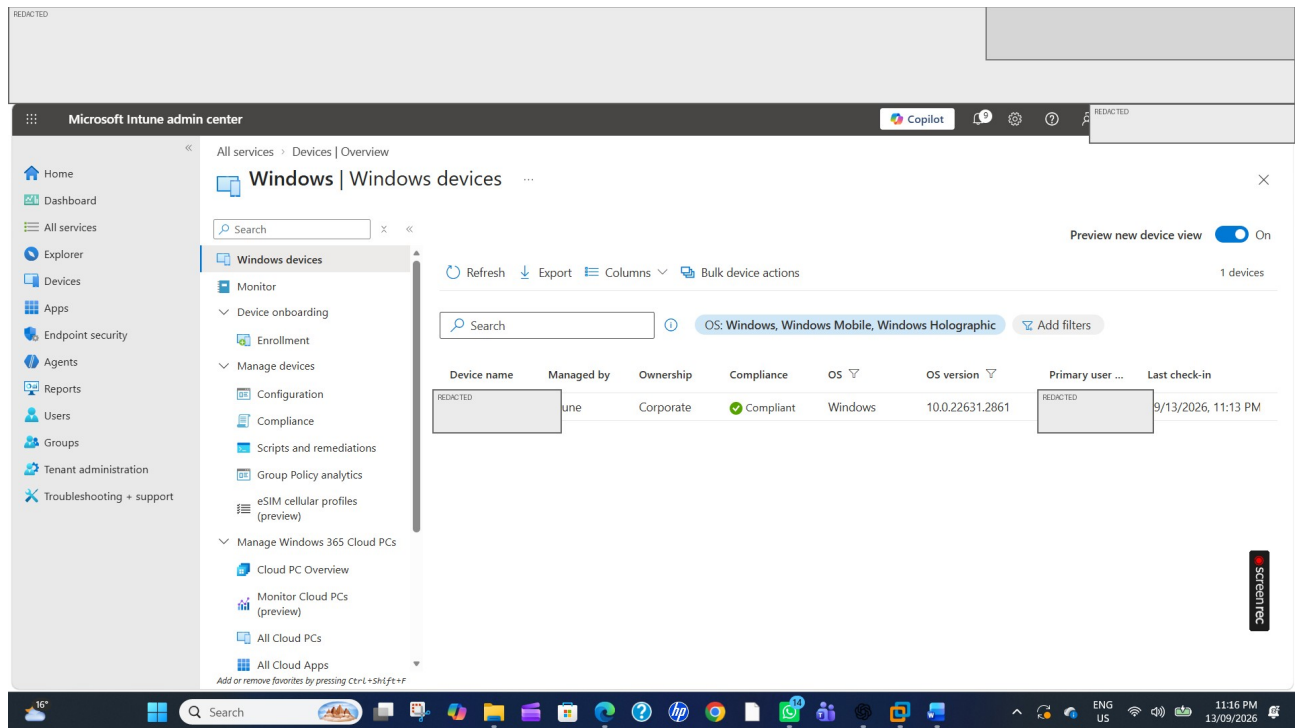


Figure 15. Final Intune verification: managed by Intune, Corporate ownership and Compliant.

14. Why enrollment was automatic even though Windows was installed manually

This is the central Autopilot concept. Windows Autopilot is a provisioning and registration technology, not a traditional operating-system imaging engine. In this pre-production test, Windows 11 Pro was installed manually first. The hardware hash was then registered with Autopilot. When the same device returned to OOB and connected to the Internet, Microsoft recognised the registered hardware identity and applied the assigned Autopilot profile.

Manual/factory Windows installation
 +
 Registered Autopilot hardware identity
 +
 Assigned deployment profile
 +
 User signs in during OOB
 =
 Microsoft Entra join + automatic Intune enrollment

In a real organisation, an OEM such as HP, Dell or Lenovo can ship Windows 11 Pro already installed and register the device for the customer. The employee can receive the laptop directly, connect it to the Internet, sign in with the work account, and allow Autopilot + Intune to provision the corporate configuration.

15. Command reference

Purpose	Command
Open PowerShell from CMD	powershell
Temporary execution policy bypass	Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process -Force
Collect Autopilot hardware hash	Get-WindowsAutopilotInfo -OutputFile C:\AutopilotHWID.csv
Verify CSV	dir C:\AutopilotHWID.csv
List volumes	Get-Volume
Copy CSV to E:	Copy-Item C:\AutopilotHWID.csv E:\
Verify copied CSV	dir E:\AutopilotHWID.csv
Inspect Autopilot diagnostics	reg Query HKLM\SOFTWARE\Microsoft\Provisioning\Diagnostics\Autopilot/s
Basic DNS/ICMP test	ping ztd.dds.microsoft.com
Test Autopilot HTTPS connectivity	Test-NetConnection ztd.dds.microsoft.com -Port 443
Verify Entra join state	dsregcmd /status
Dynamic Autopilot device group rule	(device.devicePhysicalIDs -any (_ -startsWith "[ZTDid]"))

16. Troubleshooting checklist

- Device shows Not assigned: confirm group membership and profile assignment, then Sync/Refresh and allow cloud processing time.
- Generic work/school OOB appears: do not assume Autopilot has loaded. Verify the Autopilot device shows Assigned in Intune and inspect diagnostics.
- Ping to ztd.dds.microsoft.com fails: test TCP 443 with Test-NetConnection instead of relying on ICMP.
- Autopilot CSV cannot be generated: confirm the Get-WindowsAutopilotInfo script/module is available and use a temporary Process-scope execution policy bypass if script execution is blocked.
- Device reaches desktop but is not in Intune: verify the user has an Intune license, MDM enrollment scope, enrollment restrictions and device limits.
- AzureAdJoined is NO: confirm the deployment profile join type, user authentication, Internet access and Entra device registration state.
- Compliance is not immediately evaluated: allow time for policy sync and check assigned compliance policies.

Autopilot vs imaging: Autopilot does not lay down the Windows operating system image. It transforms an existing/factory Windows installation into an organisation-managed device by applying cloud provisioning, identity, MDM, policy and application configuration.

17. Build out the Autopilot pre-production management stack

After the base Autopilot workflow was validated, the same pre-production environment was extended beyond enrollment. The objective was to test the controls that turn an enrolled Windows 11 endpoint

into a managed corporate endpoint. The build therefore included an Autopilot deployment profile, Enrollment Status Page (ESP), Windows compliance policy, device configuration/security policies, Windows LAPS and BitLocker disk-encryption policy.

- Autopilot deployment profile - controls the user-driven OOB and Microsoft Entra join experience.
- Enrollment Status Page (ESP) - provides provisioning status during initial device setup and can hold the user at setup while required device configuration is applied.
- Compliance policy - evaluates whether the Windows endpoint meets the organisation's required compliance state.
- Configuration and endpoint-security policies - apply post-enrollment management and security controls to the Autopilot device.
- Windows LAPS - manages and rotates the local Administrator password and backs it up to Microsoft Entra ID.
- BitLocker - provides full-volume encryption with recovery-key escrow for administrative recovery.

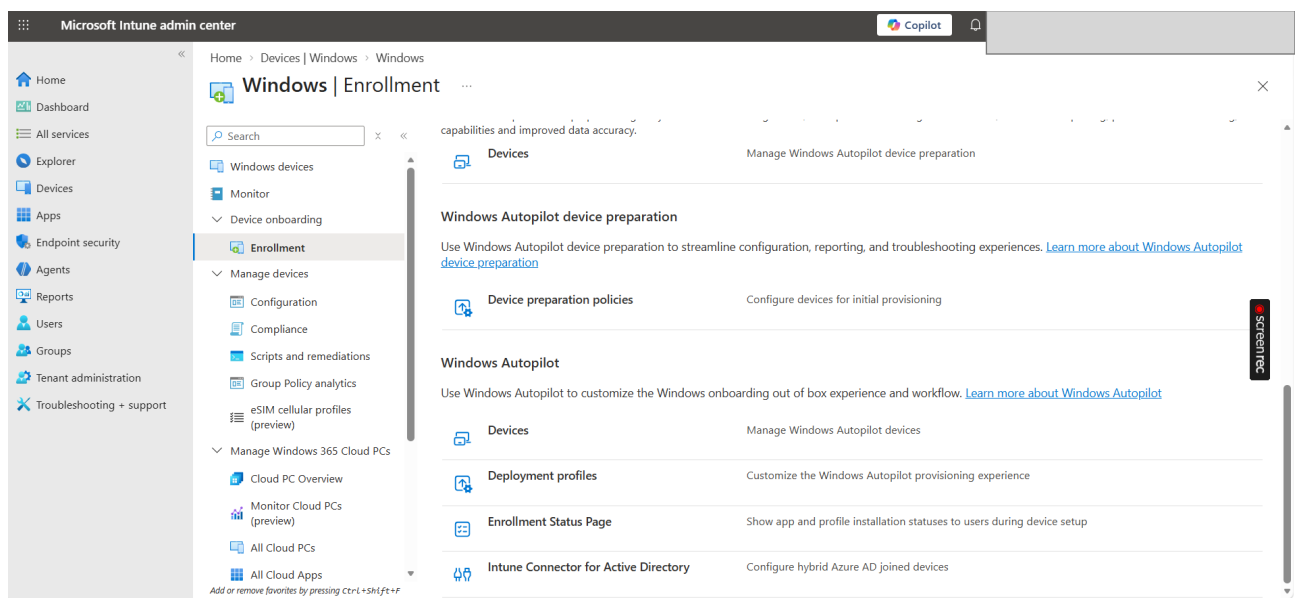


Figure 16. Windows enrollment workspace showing the Autopilot deployment and Enrollment Status Page components used in the pre-production build.

18. Enrollment Status Page (ESP)

An Enrollment Status Page was included in the Autopilot design so the provisioning workflow could be validated as a controlled corporate build rather than only an Entra join. ESP is part of the Windows enrollment workflow and is used to surface device setup progress while assigned policies and applications are being processed. This makes the test closer to an enterprise deployment experience, where the device should reach the desktop only after the required provisioning stages have been evaluated.

Validation focus: confirm that the Autopilot device enters the managed enrollment workflow, receives the assigned configuration, and reports setup progress before the user begins normal use.

19. Windows compliance policy

A Windows 10/11 compliance policy named Windows-PreProd-Compliance was created for the pre-production device scope. Compliance adds a health/governance layer after enrollment: Autopilot provisions the device, Intune manages it, and the compliance policy evaluates whether the endpoint satisfies the required state. The final Autopilot VM subsequently reported Compliant in Intune.

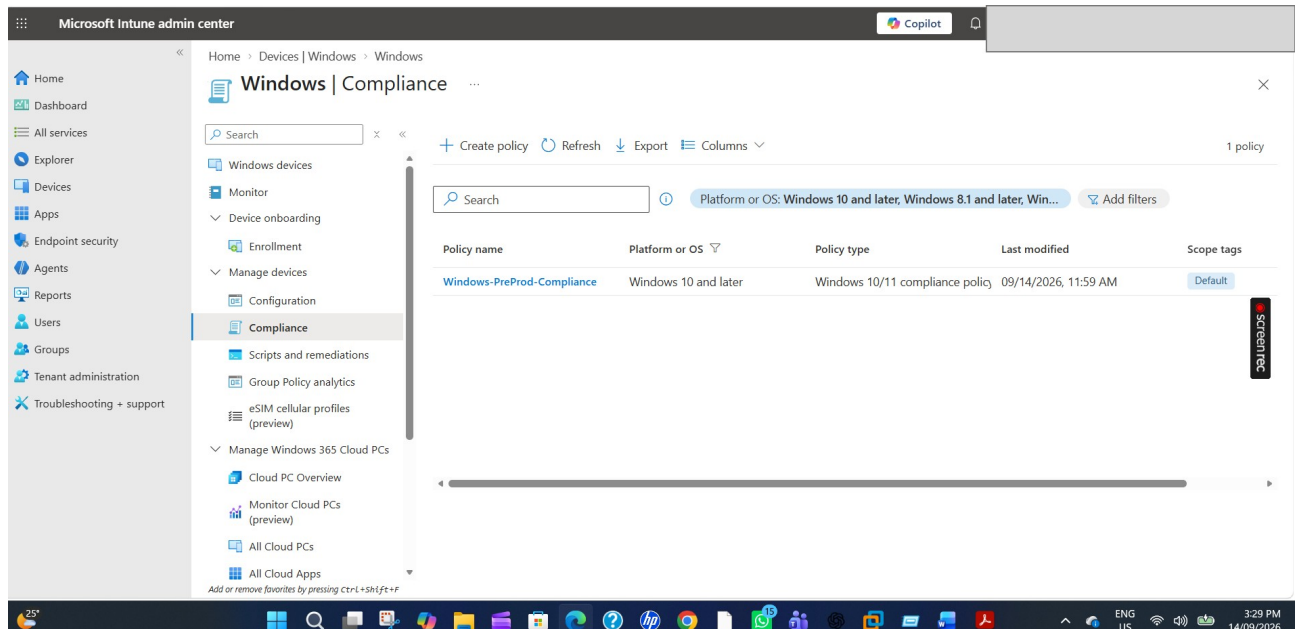


Figure 17. Windows-PreProd-Compliance policy in Microsoft Intune.

20. Configuration and endpoint-security policies

The pre-production build also used Windows configuration/security policy targeting so the Autopilot endpoint would receive security controls automatically after enrollment. The Intune Windows Configuration view shows the Windows-PreProd-BitLocker and Windows-PreProd-LAPS policies as part of the managed Windows policy set. This demonstrates the intended enterprise model: provisioning and security configuration are separate layers, but they converge on the same managed endpoint.

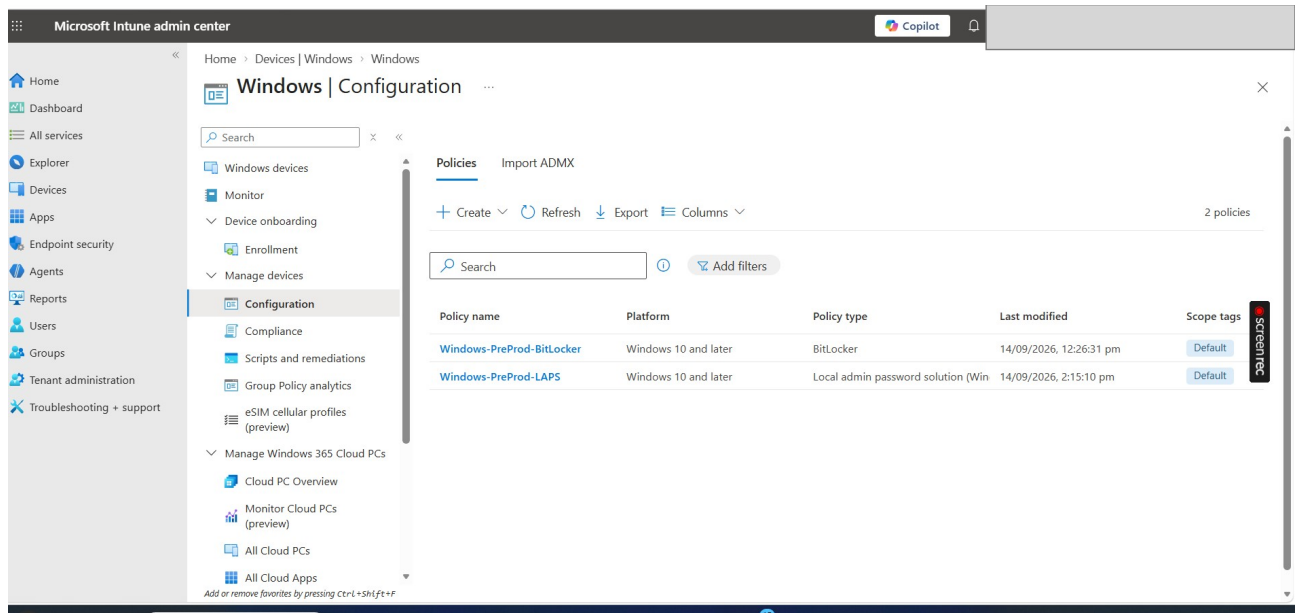


Figure 18. Windows Configuration policy view showing the pre-production BitLocker and LAPS policies.

21. BitLocker disk-encryption policy

A BitLocker policy named Windows-PreProd-BitLocker was created and assigned through Intune Endpoint security > Disk encryption. Adding BitLocker to the Autopilot build validates the disk-encryption layer expected on a corporate Windows endpoint. In an enterprise deployment, BitLocker protects data at rest and recovery-key escrow provides an administrative recovery path when a device enters recovery mode.

The policy is shown as Assigned in the Intune disk-encryption workspace. This policy is part of the same pre-production endpoint-management stack as Autopilot, ESP, compliance and LAPS.

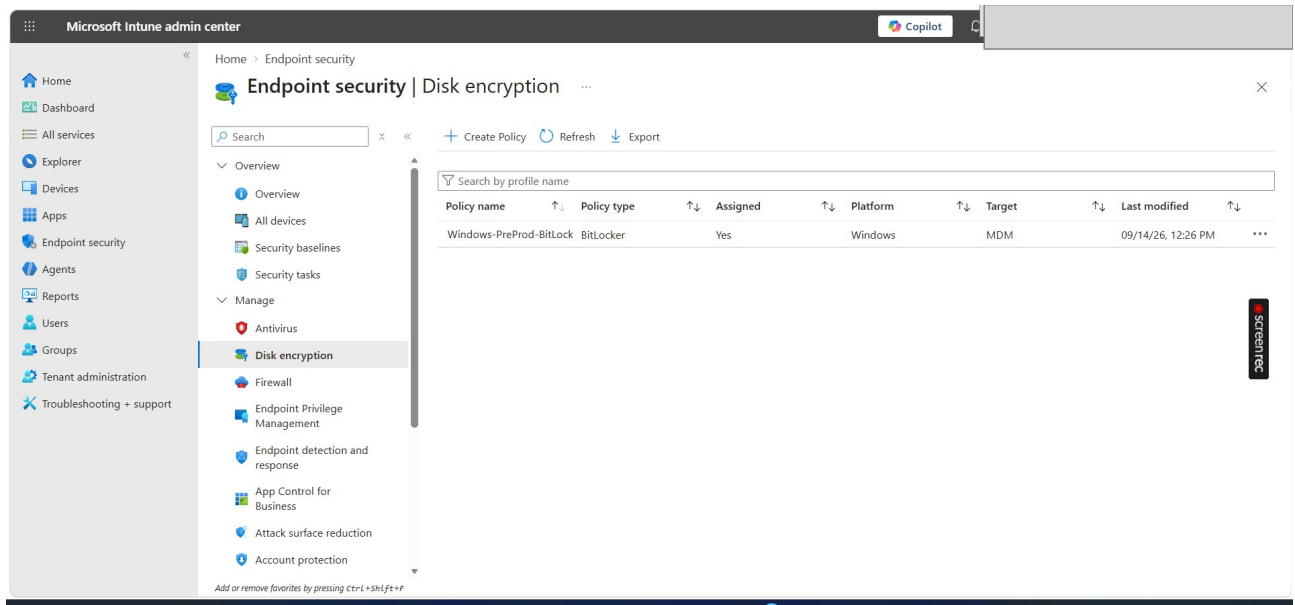


Figure 19. Windows-PreProd-BitLocker policy assigned from Endpoint security > Disk encryption.

22. Extend the Autopilot build with Windows LAPS

After validating Autopilot registration, Microsoft Entra join and automatic Intune enrollment, the pre-production build was extended with Windows Local Administrator Password Solution (Windows LAPS). This adds an endpoint security control to the same cloud-managed device lifecycle rather than leaving a shared or static local administrator password on managed Windows devices.

Security objective: each managed device receives its own strong local administrator password. Windows rotates that password automatically and backs it up to Microsoft Entra ID for authorised recovery.

Why LAPS was added to this project:

- Eliminate reuse of the same local administrator password across multiple endpoints.
- Reduce lateral-movement risk if one local administrator credential is exposed.
- Allow authorised IT administrators to recover a device-specific local password from Microsoft Entra ID when support or recovery access is required.
- Automatically rotate the password instead of relying on manual password changes.
- Validate that endpoint security configuration can be layered onto the Autopilot/Intune build before enterprise rollout.

23. Create the Windows LAPS policy in Intune

In the Microsoft Intune admin center, navigate to Endpoint security > Account protection and create a Local admin password solution (Windows LAPS) policy. The policy created in this pre-production environment was named Windows-PreProd-LAPS.

Setting	Pre-production value
Backup directory	Back up the password to Microsoft Entra ID only
Password age	30 days

Setting	Pre-production value
Administrator account name	Not configured (built-in Administrator targeted)
Password complexity	Large + small letters + numbers + special characters
Password length	14
Post-authentication action	Reset the password and log off the managed account after grace period
Post-authentication reset delay	8 hours
Automatic account management	Enabled
Automatic account management target	Manage a new/custom administrator account setting as configured in policy
Enable account	Enabled

Home > Endpoint security | Account protection

Create Policy

Targeting:

Password Age Days ⓘ ☒ Configured

Administrator Account Name ⓘ ☐ Not Configured

Password Complexity ⓘ

Password Length ⓘ ☒ Configured

Post Authentication Actions ⓘ

Post Authentication Reset Delay ⓘ ☐ Not Configured

Automatic Account Management Enabled ⓘ

Figure 20. Windows LAPS policy configuration in Intune.

The policy was assigned to the same pre-production device targeting scope used for the Autopilot-managed endpoint. This means additional devices entering the targeted device group can receive the LAPS configuration without creating a separate policy for each VM or physical computer.

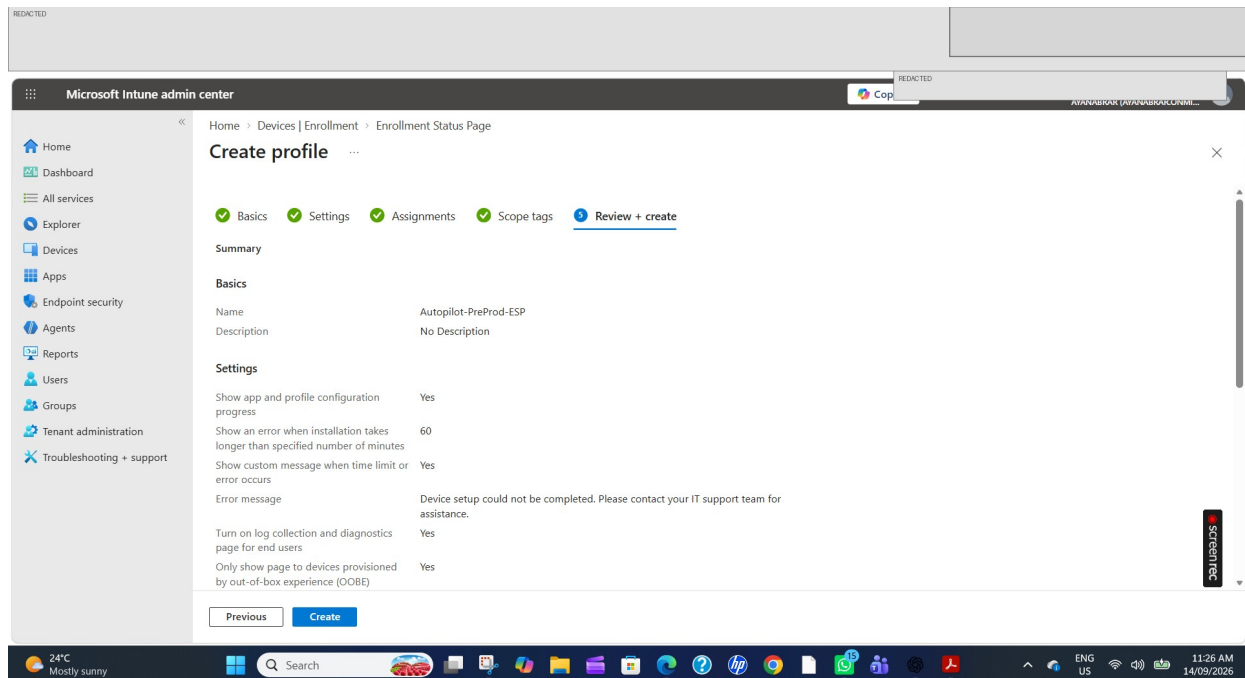


Figure 21. Windows-PreProd-LAPS policy showing successful assignment status.

24. Initial LAPS validation and troubleshooting

The first LAPS validation did not immediately succeed. PowerShell event-log review showed that LAPS policy processing was occurring, but password backup to Microsoft Entra ID failed. This was treated as a useful pre-production troubleshooting finding rather than hidden from the validation record.

```
Get-WinEvent -LogName "Microsoft-Windows-LAPS/Operational" -MaxEvents 30 | Format-List
TimeCreated,Id,LevelDisplayName,Message

Invoke-LapsPolicyProcessing

Get-LapsDiagnostics
```

The key failure recorded in the Windows LAPS Operational log was an HTTP 400 response stating that the Local admin password solution was not enabled for the tenant. This explained why the Intune policy could reach the endpoint while Microsoft Entra ID rejected the password backup operation.

```
Microsoft Windows [Version 10.0.22631.2861]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32\powershell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\System32> Get-WinEvent -LogName "Microsoft-Windows-BitLocker-API/Management" -MaxEvents 20 | Format-Table TimeCreated,Id,LevelDisplayName,Message -Wrap
Get-WinEvent : There is not an event log on the localhost computer that matches "Microsoft-Windows-BitLocker-API/Management".
At line:1 char:1
+ Get-WinEvent -LogName "Microsoft-Windows-BitLocker-API/Management" -M ...
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (Microsoft-Windo...-API/Management:String) [Get-WinEvent], Exception
+ FullyQualifiedErrorId : NoMatchingLogsFound,Microsoft.PowerShell.Commands.GetWinEventCommand

PS C:\Windows\System32> Get-WinEvent -ListLog *BitLocker* | Select-Object LogName, IsEnabled, RecordCount

LogName                                     IsEnabled RecordCount
-----
Microsoft-Windows-BitLocker/BitLocker Operational      False
Microsoft-Windows-BitLocker/BitLocker Management       True    26
Microsoft-Windows-BitLocker-DrivePreparationTool/Operational True     0
Microsoft-Windows-BitLocker-DrivePreparationTool/Admin  True     0

PS C:\Windows\System32>
```

Figure 22. LAPS Operational log identifying the tenant-level enablement problem.

Troubleshooting lesson: a policy can show as assigned/succeeded in Intune while a dependent cloud service setting still prevents the full end-to-end function. Validate both policy delivery and the service-side result.

25. Enable Windows LAPS in Microsoft Entra ID

In the Microsoft Entra admin center, navigate to Entra ID > Devices > Device settings. Under Local administrator settings, set Enable Microsoft Entra Local Administrator Password Solution (LAPS) to Yes and save the change. This tenant-level setting permits eligible devices to back up their Windows LAPS passwords to Microsoft Entra ID.

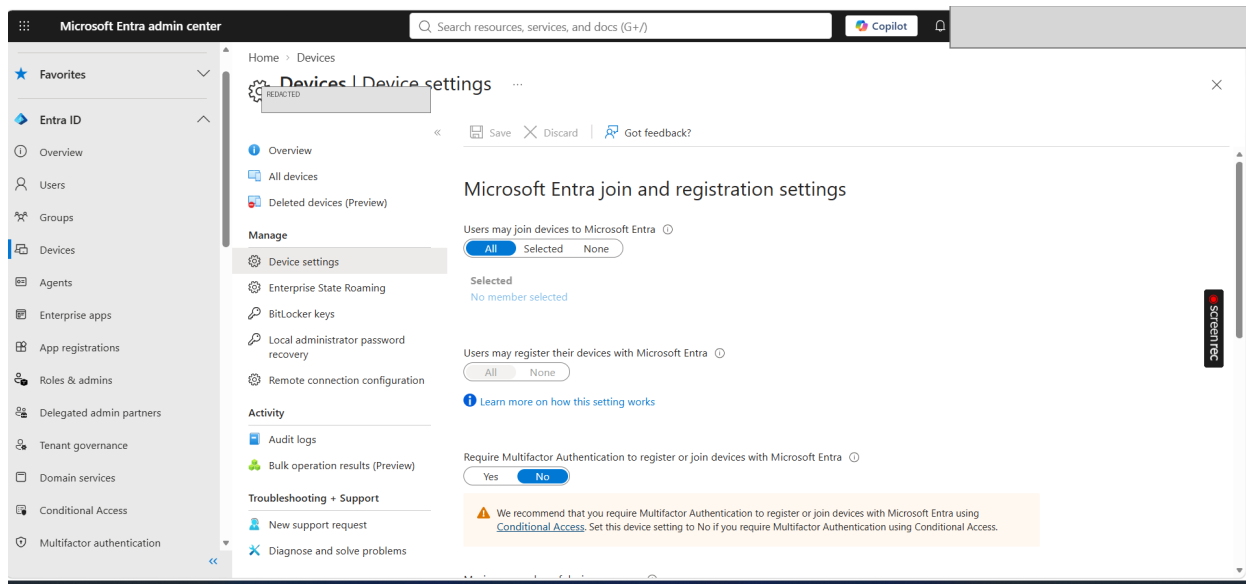


Figure 23. Microsoft Entra device settings used to enable Windows LAPS for the tenant.

26. Force processing and confirm successful password backup

After enabling LAPS in Microsoft Entra ID, the policy was processed again on the Autopilot VM. The Windows LAPS Operational log then confirmed the complete workflow: local password update, Microsoft Entra password backup, and successful policy processing.

```
Invoke-LapsPolicyProcessing
```

```
Get-WinEvent -LogName "Microsoft-Windows-LAPS/Operational" -MaxEvents 10 |  
Select-Object TimeCreated,Id,LevelDisplayName,Message |  
Format-List
```

- Event 10004: LAPS policy processing succeeded.
- Event 10020: LAPS successfully updated the local admin account with the new password.
- Event 10029: LAPS successfully updated Azure Active Directory / Microsoft Entra ID with the new password.

```
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.  
TimeCreated      : 9/13/2026 6:22:29 AM  
Id               : 10002  
LevelDisplayName : Information  
Message          : The Local Administrator Password dll was unloaded.  
  
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.  
TimeCreated      : 9/13/2026 6:19:50 AM  
Id               : 10004  
LevelDisplayName : Information  
Message          : LAPS policy processing succeeded.  
  
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.  
TimeCreated      : 9/13/2026 6:19:50 AM  
Id               : 10024  
LevelDisplayName : Information  
Message          : LAPS policy is configured as disabled.  
  
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.  
TimeCreated      : 9/13/2026 6:19:50 AM  
Id               : 10003  
LevelDisplayName : Information  
Message          : LAPS policy processing is now starting.  
  
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.  
TimeCreated      : 9/13/2026 6:09:50 AM  
Id               : 10000  
LevelDisplayName : Information  
Message          : The Local Administrator Password feature was successfully loaded and initialized.  
  
See https://go.microsoft.com/fwlink/?linkid=2220550 for more information.  
  
PS C:\Windows\System32>  
PS C:\Windows\System32>  
PS C:\Windows\System32> Invoke-LapsPolicyProcessing  
Invoke-LapsPolicyProcessing : 2147942800 failed with hr:0x80070190  
At line:1 char:1  
+ Invoke-LapsPolicyProcessing  
+ ~~~~~  
+ CategoryInfo          : InvalidOperation: (:) [Invoke-LapsPolicyProcessing], LapsPowershellException  
+ FullyQualifiedErrorId : 2147942800 failed with hr:0x80070190,Microsoft.Windows.LAPS.InvokeLapsPolicyProcessing  
  
PS C:\Windows\System32>
```

Figure 24. Successful LAPS processing and password backup events on the Autopilot VM.

27. Recover and use the LAPS-managed local administrator credential

In Microsoft Entra Local administrator password recovery, the validated VM appeared with the account name Administrator, a last password rotation timestamp and a next password rotation timestamp. The recovery password itself is not exposed in this public document.

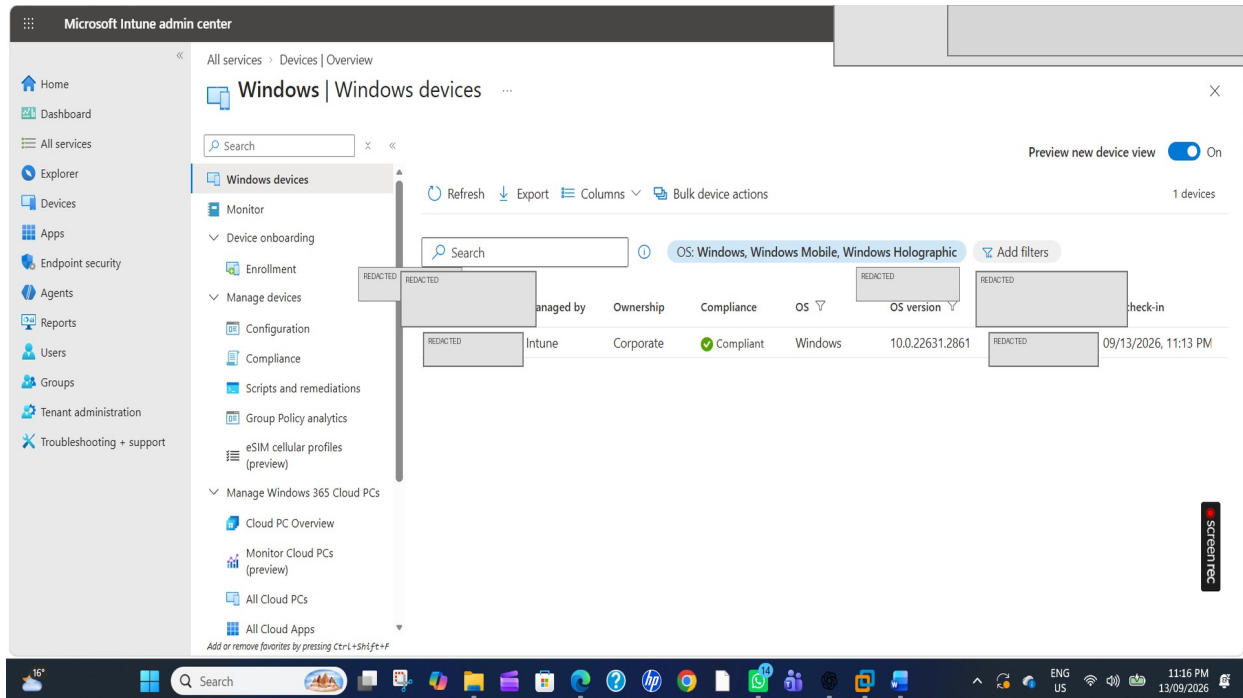


Figure 25. Intune device record used during Windows LAPS validation; device and user identifiers are redacted for public release.

For a local sign-in test, use the local account syntax below. The `.\` prefix tells Windows to authenticate against the local computer rather than Microsoft Entra ID.

Username: `.\Administrator`

Password: `<current password retrieved from Microsoft Entra LAPS>`

Before testing, the local account state can be confirmed from an elevated PowerShell session:

```
Get-LocalUser -Name Administrator
```

Operational principle: administrators do not need to know or reuse a permanent local admin password. They retrieve the current device-specific credential only when required, and LAPS rotates it according to policy.

28. How LAPS scales to additional Autopilot VMs and enterprise devices

A separate LAPS policy is not required for every new VM. When another Windows 11 device is registered with Autopilot, becomes a member of the targeted dynamic device group, completes Microsoft Entra join and enrolls into Intune, the existing Windows-PreProd-LAPS policy can apply automatically if the new device is within the assignment scope.

New Windows device

- > Autopilot registration
- > Dynamic device group
- > Autopilot profile

- > Microsoft Entra join
- > Intune enrollment
- > Windows-PreProd-LAPS policy
- > Unique local admin password
- > Password backed up to Microsoft Entra ID

Each device receives a different LAPS-managed password. The password displayed for one device must not be expected to work on another device. This per-device credential model is one of the main security benefits of LAPS.

29. Updated command reference - Windows LAPS

Purpose	Command
List local accounts	Get-LocalUser
Check built-in Administrator state	Get-LocalUser -Name Administrator
Force LAPS policy processing	Invoke-LapsPolicyProcessing
Collect LAPS diagnostic bundle	Get-LapsDiagnostics
Review recent LAPS events	Get-WinEvent -LogName "Microsoft-Windows-LAPS/Operational" -MaxEvents 10 Format-List TimeCreated,Id,LevelDisplayName,Message
Local LAPS sign-in username	.\Administrator

30. End-to-end pre-production architecture

The final validated workflow can be represented as:

Windows 11 device / VMware VM -> Hardware hash registration -> Dynamic Autopilot device group -> Autopilot deployment profile -> OOB -> Microsoft Entra join -> Automatic Intune enrollment -> Enrollment Status Page -> Compliance evaluation -> Configuration & endpoint-security policies -> BitLocker encryption -> Windows LAPS password rotation and Entra backup -> Managed corporate endpoint

31. Final pre-production validation outcome

The completed pre-production project validates a layered Windows Autopilot deployment rather than enrollment alone. The Windows 11 VMware endpoint was registered with Windows Autopilot, dynamically targeted, assigned a user-driven deployment profile, Microsoft Entra joined and automatically enrolled into Intune. The build was then extended with Enrollment Status Page, Windows compliance, configuration/security policies, BitLocker disk encryption and Windows LAPS. LAPS troubleshooting also validated tenant-level enablement, policy processing, password rotation and successful password backup to Microsoft Entra ID. Together these controls demonstrate the workflow from initial provisioning through ongoing endpoint security and administrative recovery.

Validation area	Result
Autopilot hardware registration	Validated
Dynamic device targeting	Validated

Validation area	Result
User-driven Autopilot profile	Validated
Microsoft Entra join	Validated
Automatic Intune enrollment	Validated
Corporate ownership / compliance visibility	Validated
Windows LAPS policy delivery	Validated
Microsoft Entra LAPS tenant enablement	Validated
Local admin password rotation	Validated
Password backup and recovery from Entra	Validated
Repeatable targeting for additional devices	Ready for further scale testing

This remains a controlled pre-production implementation. It demonstrates the technical workflow and troubleshooting process, but enterprise rollout should still include formal change control, least-privilege role design for password recovery, pilot rings, device naming standards, security baselines, application deployment, update management, reporting, rollback/recovery procedures and documented operational ownership.